

Math 250, Thursday, Week 3

DZB

September 2, 2021

“Tips” on mathematical writing.

The intent with negations is that you ‘distrubute’ the negation ‘all the way through’.

Technically, you can negate any sentence by adding ”it is not true that” to the beginning.

All triangles are equilateral

some triangles are not equilateral.

‘This sentence has five words’ “It is not true that this sentence has five words”

Proof by contradiction. Here is a template.

- We have a statement P that we want to prove.
- There are ‘2 cases’: either P is true or P is false.
- If we can rule out the case when P is false, then P is true.
- To begin, we assume that P is false. (I.e., we assume $\neg P$.)
- ‘Argue’; i.e., we exhibit some (correct) chain of reasoning, and end up with some statement Q .
- (I.e., we write out a proof of $\neg P \Rightarrow Q$.)

- We observe (or give a proof that) Q is actually false.
- we conclude that $\neg P$ is false.
- Thus P is true.

How do you know when to proceed by contradiction? Use contradiction when the negation ‘gives you something to work with’.

Hieroglyph: $((\neg P \Rightarrow Q) \wedge \neg Q) \Rightarrow P$ (this is a statement form, and this is something that you can prove to be true by writing out a truth table).

Definition. We say that a number x is rational if there exist integers a, b such that $b \neq 0$ and $x = a/b$. We write $\mathbf{Q}$ for rational numbers. We say that x is irrational if $x \notin \mathbf{Q}$.

Examples: $1/3 \in \mathbf{Q}$; $\pi \notin \mathbf{Q}$; e is not rational; $4/6$ is rational; $4/6 = 2/3$. We say that a/b is reduced if a and b have no common prime factors, and $a \geq 0$. (Example: $-3/5 = 3/(-5)$.)

Proposition: $\sqrt{2}$ is irrational.

Sidebar: what does it mean to say that $\sqrt{2} \notin \mathbf{Q}$? I.e., what is the negation of the definition of rational? For all $a, b \in \mathbf{Z}$ such that $b \neq 0$, $x \neq a/b$.

We want to show that $\sqrt{2} = a/b$ has no solutions (with $a, b \in \mathbf{Z}$).

Infinitude of primes.

Definition: a positive integer $p > 1$ is prime if the only positive divisors are 1 and p . A positive integer $n > 2$ is composite if it is not prime, i.e., there exists d such $d \mid n$ and $1 < d < n$. Equivalently, n is composite if we can write $n = ab$, where a and b are positive integers, and neither is equal to 1 or n .

Theorem (Euclid): there are infinitely many primes.

Proof.